National College of Ireland

School of Computing

Assessment Submission Cover Page

Module title:	Deep Learning and Generative AI
Module code:	H9DLGA
Assessment title:	Repeat TABA (Project + Paper Review)
Assessment weighting:	100% of module marks
Lecturers:	A.Sahni / A.Staikopoulos / M.Garg / A.Fajemisin
Student name:	Marco Aurélio Minozzo
Student number:	x24153362
Programme:	MSc Data Analytics
Assessment type:	Individual repeat project

Declaration

I confirm that this assignment is my own work. Where I have used the work of others, this is acknowledged. I understand that the School of Computing will screen submissions for academic misconduct (plagiarism and collusion) via Turnitin.

Signature: _____ Date: _____

Documents included in this PDF

1. Assessment cover page (this page)
2. IEEE conference-format project report
3. Distribution of work (individual)
4. Question 2 paper review will be appended when complete (Elsevier-style review of *LLM Collaboration with Multi-Agent Reinforcement Learning*; student number ends in 2)

Distribution of Work

Student: Marco Aurélio Minozzo **Number:** x24153362
Assessment: Individual — all components completed by the sole author.

Component	Contribution
Research design	Research question, CRISP-DM plan, Galaxy10 DECaLS selection
Literature	Related work using module supporting materials and cited astronomy/DL sources
Data pipeline	Download, stratified splits, imbalance analysis
Generative model	ConvVAE training, latent sampling, synthetic balancing
Discriminative model	ResNet50 transfer learning (real-only vs real+VAE)
Evaluation	Metrics, confusion matrices, minority-class comparison
IEEE report	Conference-style paper
Code package	Commented source under `src/` and notebook

Work undertaken that may not appear in the final paper: exploratory smoke tests on synthetic subsets; hyperparameter trials for VAE epochs/latent dimension prior to the reported configuration; visualisation checks of reconstructions.